Exemplars Move Retrieval, Instructions Move Rejection: An Anatomy of Inference-Time Interventions for Closed-Book Knowledge Base Construction

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Abstract

We describe our system for the AKBC 2026 shared task on closed-book knowledge base construction. Our final system is a single pretrained-only backbone, Gemma-3-27B-pt, queried entirely in raw text-completion registers with 20 candidates per query, gated relation-specific aggregation, and no arbiter; it uses 27.4B of the 32B parameter budget and scores a row-averaged Macro F1 of 0.6670 on the official test set (6th on the leaderboard), against 0.313 for the official baseline. Its predecessor—two deliberately different prompts over Mistral Small 3.2 24B with a cross-family 8B arbiter, 31.8B in total—scores 0.6247. Most of the paper presents a mechanism study over three backbones and roughly forty controlled experiments, organized around three results. First, how reliably a backbone recalls the value stored for an entity label dominated every intervention we tried: each backbone change moved the score more than all our prompt, sampling, and selection experiments combined, and the winner is a pretrained checkpoint: its instruction-tuned sibling, queried in the same registers so that only the weights differ, loses 0.11 on validation. Second, exemplars move retrieval and instruction text does not—two in-prompt exemplars demonstrating a verbalized recall move the answer mode by +6 validation rows, while the same request as an instruction moves nothing, and replacing synthetic exemplars with real training answers adds a further +2.1 points—but what instructions do move is *rejection*: a register that writes a one-line biography before naming a city lets the model decline rows it cannot answer, raising that relation from 0.55 to 0.61 on validation and reproducing on train and across seeds. Third, validation gains transferred to the hidden test set when the change was gated, when the gate consulted information from outside the model, and when the gain was stable across sampling seeds; each criterion has a counterexample that cost us test rows.

1 Introduction

The LM-KBC shared task series (Singhania et al., 2023; Kalo et al., 2024) probes how much factual knowledge language models can turn into knowledge base entries: given a subject and a relation, a system must output *all* correct object values—possibly none, one, or hundreds—under closed-book conditions: no retrieval, no external corpora, no fine-tuning, and at most 32B neural parameters at inference.

Under these conditions the only controls are at inference time: model choice, prompts, sampling, and aggregation over samples. Our development proceeded in three stages: a Qwen3.5-27B thinking system (test 0.571), a Mistral Small 3.2 system with a second prompt and a cross-family 8B arbiter (0.6247), and a single pretrained-only Gemma-3-27B backbone in completion registers (0.6670). Alongside the system we report the controlled experiments behind each decision. Those experiments are measured on validation and, where a threshold had to be chosen, on the training split; the hidden test set enters only as the outcome of submissions, which lets us ask separately which validation gains survived it.

1. **Backbone recall reliability dominates.** A 12-row probe (30 GPU-minutes per model) ranked backbones consistently with downstream quality in every deployment decision we evaluated, and both backbone changes beat all our prompt and selection work combined (Sec. 4.1).

2. **Answers commit early** to the first concrete value they generate (Sec. 4.2), across backbones.

3. **Exemplar style moves retrieval; instruction text does not** (Sec. 4.3); instructions move rejection. The effect is conditional on recall reliability and on post-training (Sec. 4.5).

4. **Within one model we found no usable selection signal**; across prompts and families we did (Sec. 4.6).

5. **Validation gains transfer when the change is gated and the gate looks outside the model** (Sec. 4.7).

6. **The residual errors are consistent with incorrect values learned from training data**, and are associated with the instruct-plus-chat setting (Sec. 4.8).

2 Task and Constraints

The 2026 task covers six relations over 478 validation and 477 test subject–relation pairs: two set-valued relations that may be empty (`countryLandBordersCountry`, `companyTradesAtStockExchange`), one single-valued string relation that may be empty (`personHasCityOfDeath`), one large-set relation (`awardWonBy`), and two numeric relations (`hasArea`, `hasCapacity`) scored at 5% relative tolerance. Gold values come from Wikidata (areas `P2046`, capacities `P1083`), with alias lists accepted for strings. We report the official "All Relations" score throughout: a Macro F1 averaged over *rows*, not relations, so the four 100-row relations dominate it. Systems are closed-book and capped at 32B inference parameters.

3 System Description

3.1 Final system

The final system generates with Gemma-3-27B-pt (the pretrained checkpoint, Q8_0, served by `llama.cpp` at a pinned revision), one model and no arbiter, 27.4B of the 32B budget. Every relation is queried in a *raw text-completion register*—a record- or infobox-style prefix that the model continues, with no chat template and no instruction turn—with 20 candidates per query at temperature 0.6, top-p 0.95 and deterministic per-candidate seeds, as in the earlier systems. Two registers depart from a bare record. `hasArea` and `hasCapacity` carry four fixed exemplars drawn from the *training* split rather than written by us (Sec. 4.3); `personHasCityOfDeath` writes a one-line biography before the answer field (Sec. 4.4). Neither changes generation or aggregation otherwise.

Aggregation is relation-specific and, in every case, *gated*: a rule applies only where an explicit support condition holds and falls back to a default otherwise. Before voting, string surfaces are folded onto public gold entity classes with an alias graph built from the released train and validation answers, using English-label matches only (validation $0.658 \rightarrow 0.663$, driven by company). Numeric candidates are clustered at 5% relative tolerance and `dominant_cluster` predicts the median of the best-supported cluster. `personHasCityOfDeath` takes a surface majority with a minimum-vote gate of 14 of 20, predicting empty below it. `companyTradesAtStockExchange` folds surface variants—case variants and acronyms that expand to a co-occurring long form (NYSE $\rightarrow$ New York Stock Exchange)—into a single count, repairing vote splits that would drop majority answers below its frequency threshold (≥ 0.5), and abstains on an empty majority. `countryLandBordersCountry` and `awardWonBy` use recall-oriented thresholds (≥ 0.3, ≥ 0.1) with the same abstention.

Reproducibility. Our submissions appear on the official Codabench leaderboard under the team name `acro-sasaki`. Code, configurations, builders, the step-by-step recipe (`docs/V4_REPRODUCE.md`), and per-experiment records are at https://github.com/acrosasaki-rgb/llm-knowledge-task-2026. Model revisions, container digests, and the dataset commit are pinned; each prediction file is a deterministic function of persisted candidate sets and a builder script (SHA-256 verified).

3.2 Predecessor system

Most experiments in Sec. 4 come from the Mistral-era system that preceded it. It samples 20 candidates per prompt from Mistral Small 3.2 24B under *two* prompts of deliberately different style for `hasArea`: a **recall prompt** with no few-shot examples but two in-prompt exemplars demonstrating a labeled recall format (`Entity:`, `Memory:` "I recall that the Wikipedia infobox lists X's total area as Y", `Unit:`, `Answer:`), and a **grounding prompt** with five bare-JSON few-shot examples naming the recall target (Wikidata `P2046`/`P1083`). When the two disagree by more than 5%, one aggregated arbiter decision—the mode over 20 samples of an unrelated-family Llama-3.1-8B—breaks the tie, switching only if the arbiter matches the secondary answer and not the primary one. The arbiter never introduces a value neither prompt proposed, but an incorrect switch can still lose a row. Total: $24\text{B} + 8\text{B} \approx 31.8\text{B}$.

System	cap	test
Baseline (Qwen3.5-9B)	✓	—
Qwen3.5-27B thinking	✓	0.571
Mistral 3.2 grounding	✓	0.595
v3: + recall prompt, area arbitration, merge	✓	0.604
v4: + cap. arbitration	✓	0.606
v5: + acronym merging	✓	0.610
v7: + city merge	×	0.608
v8: re-keying*	×	0.625
v10: + bridge	✓	0.623
v11: − bridge	✓	0.625
v12: Gemma-3-27B-pt	✓	0.649
v13: + alias graph	✓	0.648
v16: + train exemplars	✓	0.659
v18: + city vote gate	✓	0.663
v19: + biography register	✓	**0.667**

Table 1: Official test scores of our submissions (row-averaged Macro F1, three decimals). Each row is its predecessor plus the named change, so the column doubles as a sequence of isolated ablations on the hidden set (Sec. 4.7); *cap* marks configurations within the 32B limit. Validation scores are not comparable across the whole table—the August release re-keyed the gold—so they are reported per experiment in Sec. 4 instead. *Following that release, which renamed 15 test subjects and dropped two rows (477→475); v3–v7 score the renamed rows as zero. Versions differing only in variants we did not deploy (v9, v14–v15, v17) are omitted for space.

3.3 Results

Four contributions are separable on test: the Qwen→Mistral backbone change (+0.024, rest of the pipeline unchanged), the Mistral-era pipeline interventions (+0.015, v3→v5), the Mistral→Gemma-pt backbone change (+0.025, v11→v12), and the two register changes of Sec. 4.3 and Sec. 4.4 (+0.018, v13→v19). A further +0.015 came from following the dataset re-keying, which corrects what is scored rather than improving the system. Per-relation test scores for the final system are: borders 0.960, company 0.832, area 0.820, city 0.530, award 0.336, capacity 0.316. The backbone change is what moved company (0.778 → 0.832) and award (0.279 → 0.336); the register changes moved capacity (0.276 → 0.316) and city (0.490 → 0.530).

4 Anatomy of Inference-Time Interventions

All experiments change one stated condition; numeric testbeds are `hasArea` and `hasCapacity` (100 validation rows × 20 candidates per prompt). **Oracle** is the number of rows whose candidate set contains at least one within-tolerance value; the **mode** row count is what `dominant_cluster` selects. The **anchor** is the first concrete value for the queried quantity in a candidate.

Model (12-row probe)	any	mode
Mistral Small 3.2 Instruct	11	8
Mistral Small 3.1 Base	10	7
Mistral Small 3.1 Instruct	9	4
Llama-3.1-8B Instruct	7	4
Aya Expanse 32B	3	1
Qwen3-8B	1	0

Table 2: The recall probe: 12 `hasArea` rows stratifying the Qwen-era system's failures, 20 candidates each; *any* = rows with a correct candidate, *mode* = rows whose modal cluster is correct. Qwen3.5/3.6-27B, Gemma 4 31B, GLM-Z1-32B and llm-jp-4-32b (not shown) sit between Llama-8B and Aya. It costs ∼30 GPU-minutes per model.

4.1 Backbone Recall Reliability Dominates

Our Qwen-era system plateaued after nine intervention experiments (prompts, temperature, output routes, thinking budgets, judge models, log-probability confidence) whose combined deployable gain was under +0.01. A 12-row probe of its wrong rows across candidate models (Table 2) showed Mistral Small 3.2 answering 11 of the 12 somewhere in its candidate set; regenerating the pipeline on Mistral with nothing else changed moved test 0.571→0.595.

The second change was larger and reversed an earlier decision. Gemma 4 31B Instruct, queried through its chat template, had the weakest `hasArea` oracle we measured (43/100) and we closed that lineage early. Probing the *pretrained* Gemma-3-27B checkpoint in a completion format instead gave the best profile of any configuration we tested—oracle 94, mode 84 on `hasArea`, borders 0.986—and a system built on it took test 0.625 → 0.649. It also supplies values no prompt reached on Mistral: secondary stock listings that five Mistral prompt variants returned with zero votes (ČEZ on Warsaw, Questerre on Oslo, Aisin on Nagoya) appear with 7–13 votes out of 20. This was the dominant empirical factor in our experiments: closed-book quality is set by which label-to-value pairings pretraining left intact and by whether the prompting format can reach them.

`hasArea` prompt (Mistral 3.2)	Few-shot examples bare-JSON ×5	none
grounding instruction	71 (o. 89)	**41** (o. 59)
recall-format text	72 (o. 83)	**78** (o. 84–89)

Table 3: The 2×2 comparison: mode-correct rows out of 100 (oracle in parentheses). The recall-format text carries two exemplars inside the instruction, so the bottom-right cell keeps two exemplars while the top-right has *none* and collapses to 41. At fixed example style the instruction moves the result by ±1; the example style moves it by +6 / −30.

4.2 Answers Commit Early

In persisted reasoning traces of the Qwen-era system, the answer value first appears at a median 17% of the trace, in 95–98% of candidates inside a retrieval template ("Commonly cited: X"); the remaining 80% of the trace never dislodges it. The verification text that follows is consistent with post-hoc justification rather than independent checking: fabricated Wikidata Q-IDs and cross-language "confirmations" agree with whatever the anchor was, and wrong candidates carry *more* of this apparatus than correct ones (6.9 vs. 4.2 source mentions). Early commitment replicates on Gemma 4 31B and Qwen3.6; the confabulated verification does not. Extended thinking is thus an *exploration* aid: it raises the oracle where recall is unreliable (Qwen +9 rows, Gemma +7) and has no measurable effect on Mistral or Magistral.

4.3 Exemplar Style Moves Retrieval; Instruction Text Does Not

The single largest prompt effect we found—`hasArea` mode $72 \rightarrow 78$, transferring to test as $0.70 \rightarrow 0.75$—looked at first like an instruction effect: a step-structured expert prompt, adapted from a style published by another participant, beat our grounding instruction by 6 rows. The 2×2 comparison (Table 3) shows the instruction text is nearly inert. What matters is the *exemplars*: with two in-prompt exemplars demonstrating a one-sentence verbalized recall ("I recall that the Wikipedia infobox lists Luxembourg's total area as 2586.4 km^2"), 99% of candidates reproduce the verbalization and the mode moves +6; with five bare-JSON few-shot exemplars, 0% verbalize—the same procedure requested as instruction text is silently ignored—and the mode does not move. Iterating the wording (persona, fixed phrase, provenance vocabulary) changed nothing, while reproducing the labeled structure with our own labels recovered the full effect.

The final system carries this one step further. Replacing our hand-written exemplars with four answers taken verbatim from the *training* split—same register, same generation, same aggregation, only the examples change—moves validation from 0.665 to 0.686, with `hasArea` $0.84 \rightarrow 0.88$ and `hasCapacity` $0.309 \rightarrow 0.361$. The gain holds across three sampling seeds (area 0.88/0.89/0.87, capacity 0.361/0.351/0.361), and no exemplar subject occurs in validation or test.

4.4 Instructions Move Rejection

Instruction text was not inert everywhere: it moved the *discarding* of already-generated candidates. Verifying each `awardWonBy` candidate with three yes/no votes and dropping only unanimous-no candidates raised precision $0.241 \rightarrow 0.280$ and F1 $0.172 \rightarrow 0.186$; the yes votes carried no information, so the usable signal was one-sided. A digit-check step inside an otherwise inert step-by-step procedure likewise cut votes for order-of-magnitude twin values from 38 to 32 with the oracle unchanged.

The clearest case is `personHasCityOfDeath`, the one relation whose answer may legitimately be empty. Asking the register to write a one-line biography before the answer field—`About:` then `City of death:`—raises it from 0.55 to 0.61 on validation against the same aggregation, reproduces on the training split $(0.52 \rightarrow 0.55)$ and on a second seed (0.60), and beats every other register we wrote. The traces show why: the model states "he is still alive" for living subjects, converting five wrong fills into abstentions, and recalls six further correct cities. The same register applied to `hasArea`, `hasCapacity` and `companyTradesAtStockExchange` changes nothing $(0 \ldots -2)$—those relations do not ask whether to answer at all.

Style, however, only moves where the model operates, not how well it separates. Compared at matched numbers of filled rows, the biography register dominates every other register we wrote (at 61 fills, 34 correct against 31), while styles that bias toward "alive" abstain more but discard correct answers at the same rate as wrong ones. Fifteen further gates for this relation, from obituary-style vetoes to internal-activation familiarity probes, moved validation by at most two rows.

4.5 The Exemplar Effect Is Conditional

Verbalized recall is a *focusing* intervention: it sharpens the anchor distribution around the most frequently recalled value (oracle $89 \to 84$ as the mode improves), so its sign follows how reliable recall already is, in mirror image to thinking. Thinking explores, gaining $+9$ rows where recall is unreliable (Qwen) and nothing where it is reliable (Mistral); verbalized recall does the opposite, zero or negative on Qwen and on Mistral's weakest relation (`hasCapacity` oracle $60 \to 47$) but $+6$ on `hasArea`. In Magistral (Mistral's RL-tuned sibling) it disappears: the verbalized form is copied at 99.9% compliance in three different channels, yet the anchor distribution does not move in any of them (70–72), suggesting that RL post-training may decouple response style from retrieval.

4.6 We Find No Usable Within-Model Selection Signal in the Features We Tested

Within one model's candidate set, minority correct values were indistinguishable from minority wrong ones on every feature we measured (votes, dispersion, trace features, log-probabilities; base rate 8%), which is why judges, confidence weighting, and consensus schemes failed in our experiments. What did extract signal is *disagreement between prompts of different style*: on the $\sim$25 rows where the recall and grounding prompts disagree by more than 5%, one aggregated decision from an unrelated-family 8B model resolves the tie, giving `hasArea` $78 \to 80$ and `hasCapacity` $26 \to 28$ on validation; of the four `hasArea` rows it changed, three were corrected and one harmed. Controls locate the conditions: same-style prompt pairs produce too little disagreement to arbitrate; promoting minority clusters by the arbiter's mode is *negative*—an 8B cannot overrule a 24B mode, only break ties; and a weak arbiter (Ministral-8B, same family, mode accuracy 23 vs. Llama's 53 of 100) gains nothing. The isolated v3→v4 ablation confirms the transfer on test: capacity F1 $0.204 \to 0.214$.

4.7 Which Validation Gains Transfer

Our submissions give per-component test measurements (Table 4). The components that transferred share two properties: each is *gated*, touching only rows that meet an explicit condition, and each gate consults information from outside the model—an unrelated-family vote, or the gold alias structure that makes two surface forms one count. The exemplar effect is the apparent exception: it changes what the backbone retrieves rather than adjudicating between candidates.

Component	val	test	transfers
recall exemplars (§4.3)	$+6$	$+5$	✓
company folding (§3)	$+1$	$+2$	✓
8B arbitration (§4.6)	$+2$	$+1$	✓
capacity prompt (best of 6)	$+2$	-2	×
second arbiter (increment)	$+1$	0	×
tolerance rescoring	$+2$	-1	×
city two-stage merge	$+11$	-1	×

Table 4: Row-level validation and test deltas for deployed components, from the isolated ablations of Table 1. The split follows neither effect size (our largest validation gain transferred worst) nor the presence of a mechanism-level rationale (the rescoring has one).

The components that did not transfer are fits to validation row distributions: the best of six capacity configurations (winner's curse), an increment to an already-saturated arbiter, and a city pipeline whose $+11$-row gain rested on the rate at which split-vote rows happen to be correct-mode. The most instructive case is a tolerance-geometry rescoring we deployed in v9–v10. Two values whose ratio is below $1.05/0.95 \approx 1.105$ admit a single prediction correct for either, so each answer can be shifted inside its own $\pm5\%$ window to also cover a runner-up cluster. It follows from the metric geometry, has no tuned constants, and gained $+1$ to $+2$ rows across every prompt pairing and arbiter seed we tested, yet it cost a row on test (`hasArea` $0.800 \to 0.790$), because it moves *every* row rather than only rows meeting a condition; we removed it in v11. Later gated experiments failed the second criterion the same way: certificates computed from the same model's internal consistency (unit agreement, answer-blind priming) admitted confident errors and lost rows. A third predictor appeared late. When we replaced our exemplars with training answers, the same change helped `hasArea` and `hasCapacity` on all three seeds we ran and helped `personHasCityOfDeath` on one seed only (0.56/0.50/0.49); we deployed all three, and on test the two seed-stable relations transferred ($+4$ capacity rows) while the seed-unstable one cost five rows. We now treat single-seed differences below about four rows on a 100-row relation as unusable. A 100-row validation set may not reliably expose these failures; in our experiments, whether a component is gated, whether its gate looks outside the model, and whether its gain survives reseeding predicted transfer better than effect size or the presence of a mechanism-level rationale.

4.8 Residual Errors Look Like Values Learned From the Corpus

The residual errors we examined are not reasoning failures. Corfu (gold 626 km^2) is answered as 592.9 by every Mistral-family instruct configuration, 593 by Qwen3.5, and 580 by Gemma: stable, model-specific wrong values that survive every prompt. Three observations support the hypothesis that they were learned from training data. First, *turnover across generations*: Qwen3.6 repairs 7 of Qwen3.5's wrong rows and breaks 11 new ones with nothing else changed, and the union oracle reaches 89/100. Second, *unrelated training lineages solve mainstream-broken rows*: llm-jp-4-32b and Aya Expanse are far weaker overall yet each partially solves Corfu, which no mainstream-corpus model in our set solves. Third, *instruction tuning costs recall*: under raw completion Mistral 3.1 Base scores 10/12-any against its own instruct's 9, and the gap replicates on Llama-3.1-8B (mode accuracy 60 vs. 53 of 100).

The last comparison is the one our final system rests on, so we ran it without the register confound: the instruction-tuned sibling of our backbone, queried in the *same* completion registers with the same sampling and aggregation, scores 0.553 on validation against 0.665 for the pretrained checkpoint (`hasArea` 0.61 vs. 0.84, `hasCapacity` 0.24 vs. 0.31, `personHasCityOfDeath` 0.38 vs. 0.52), reproduced on two seeds. Since weights are the only difference, the loss belongs to instruction tuning itself rather than to the chat template, at least in this lineage—and the abstention behaviour that makes the biography register work (Sec. 4.4) is absent in the tuned sibling, which fabricates instead. What all of this establishes is that the accessible ceiling is a property of the checkpoint, not of prompt engineering within one setting.

5 Related Work

Probing factual knowledge in LMs goes back to LAMA (Petroni et al., 2019); the LM-KBC series (Singhania et al., 2023; Kalo et al., 2024) extends it to KB construction with unconstrained cardinalities. Self-consistency (Wang et al., 2023) aggregates sampled answers by majority vote and ReWiSe (Albert-Roulhac and Zouaq, 2025) tuned it relation-wise; we extend the line to empty answer sets, numeric tolerance, surface-form folding, and cross-family arbitration. Sprague et al. (2025) find chain-of-thought helps reasoning but not knowledge recall; we refine this for recall tasks, where exemplars move retrieval although instructions do not, and where a chain-of-thought register helps exactly one relation—the one that must decide whether to answer. Ma and Hewitt (2026) report prompting moves parametric knowledge access by 1–2% and RL fine-tuning by $\sim$10%; our exemplar effect sits between, and fine-tuning is prohibited here. Yuchi et al. (2026) find probe-recoverable numeric knowledge exceeds what models verbalize, one reading of our oracle–mode gaps; Kalyan et al. (2021) and Park et al. (2022) document the magnitude and unit fragilities our components target; and grounding instructions instantiate "According to. . . " cues (Weller et al., 2024).

6 Conclusion

Our final system is a single pretrained-only 27.4B backbone queried in completion registers with gated aggregation (test 0.6670, 6th on the leaderboard); the two-prompt, arbiter-ensembled 24B system it replaced scores 0.6247. The most reliable finding behind that progression is that backbone recall reliability, together with a prompting format that can reach it, outweighed every prompt-level intervention we tested, and that a half-hour probe ranked candidates well enough to act on.

Two rules earned their place. Choose exemplars, not instructions, to change what a model retrieves; instruction text in our experiments moved only the rejection of already-generated candidates. And before deploying what a small validation set endorses, ask whether the component is gated and whether its gate consults anything outside the model—those that failed this test cost us rows even with a clean mechanism-level rationale. Both rules are bounded by what we measured: three backbones and one task edition.

Limitations

The mechanism findings rest on two model families in depth (Qwen, Mistral), with seed replications only for the exemplar, arbitration and register results. The final backbone is the least ablated configuration here: its per-component contributions are not separated on test. The 12-row probe is stratified on one system's failures and its ranking is validated on the decisions we took, not a heldout model set. Counts on 100-row relations carry ±1–2 row noise, which bounds several negative results. The instruct-versus-pretrained control of Sec. 4.8 holds the register fixed, but our earlier Mistral and Llama comparisons do not, and the account of residual errors is observational. Thresholds are tuned on validation or train, and the transfer criterion of Sec. 4.7 comes from one task edition. Two choices used the hidden set and are therefore not validation-reproducible: after training exemplars cost five city rows on test we reverted that relation to its earlier register, and we compared six city vote thresholds on test before keeping the one reported here. The final city configuration should be read as tuned rather than predicted; every other decision was made on validation and train alone. Finally, submissions v6–v9 included a second 8B arbiter that put the ensemble at ≈39.8B, over the 32B cap; the violation was ours, found in a self-audit, and v10 removed it at no test cost. Their test numbers should be read as at-risk under the cap rule; v11 onwards are compliant.

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